

October 27, 2025

1. An electron has a kinetic energy of 2.5 eV.

- (a) What is its velocity?
- (b) What is its momentum?
- (c) What is the wavelength of the associated de Broglie wave?

Megoldás (Feladat 1). Az elektron kinetikus energiája

$$K = 2.5 \text{ eV} = 2.5 \times 1.602176634 \times 10^{-19} \text{ J} = 4.005441585 \times 10^{-19} \text{ J}.$$

(A) A sebesség nemrelativisztikus közelítéssel:

$$v = \sqrt{\frac{2K}{m_e}} = \sqrt{\frac{2 \cdot 4.005441585 \times 10^{-19}}{9.10938356 \times 10^{-31}}} \approx 9.3777 \times 10^5 \text{ m/s}.$$

(B) Az impulzus:

$$p = m_e v \approx 9.10938356 \times 10^{-31} \cdot 9.37768638 \times 10^5 \approx 8.5425 \times 10^{-25} \text{ kg} \cdot \text{m/s}.$$

(C) A de Broglie-hullámhossz:

$$\lambda = \frac{h}{p} \approx \frac{6.62607015 \times 10^{-34}}{8.5424942 \times 10^{-25}} \approx 7.7566 \times 10^{-10} \text{ m} \approx 0.776 \text{ nm}.$$

2. A metal surface is illuminated with light of wavelength $2.5 \cdot 10^{-7} \text{ m}$. What is the velocity of the emitted electrons if the photoelectric effect starts at light with a wavelength of $2.77 \cdot 10^{-7} \text{ m}$? ($m_e = 9.1 \cdot 10^{-31} \text{ kg}$)
3. A photon ejects an electron with a maximum kinetic energy of 0.64 eV from a metal for which the work function is 3.74 eV.
 - (a) What is the energy of the photon in electronvolts?
 - (b) What is the wavelength of the applied ultraviolet radiation?
4. In a Compton scattering experiment, X-rays with a wavelength of 0.104 nm are used. ($m_e = 9.1 \cdot 10^{-31} \text{ kg}$)
 - (a) At what scattering angle does the wavelength of the radiation increase by 1%?
 - (b) At what angle does the wavelength become 0.05% larger?
5. X-rays are scattered by electrons. $\lambda_0 = 10^{-11} \text{ m}$. The magnitude of the wavelength change is $2.6 \cdot 10^{-12} \text{ m}$.
 - (a) What is the scattering angle of the photons?
 - (b) By how much did the photon energy change during the process? ($m_e = 9.1 \cdot 10^{-31} \text{ kg}$)